



Water superabsorbent composites of dialdehyde cellulose from various cellulosic sources: a comparative study

Youssef Chaab · Hicham Aitbella · Ahmed Yassine Boussif ·
Mohammed Bezbiz · Meriem Latifi · Larbi Belachemi · Hicham Ben Youcef ·
Céline Moreau · Bernard Cathala · Hamid Kaddami

Received: 12 October 2024 / Accepted: 21 June 2025 / Published online: 10 July 2025
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract In the context of global water shortages, there is an urgent need to implement effective strategies for the management of water resources. The objective of our research is to synthesize superabsorbent polymers to efficiently utilize water resources in agriculture, with a particular focus on the impact of cellulose sources and properties on absorption capacity. A hybrid superabsorbant polymer (SAP) hydrogel was prepared by combining 5% (wt) cellulose pulps or their dialdehyde cellulose (DACs) with a poly (acrylic co-itaconic) copolymer acid through radical chain polymerization. The crosslinker used was *N,N*-methylene-bis-acrylamide (MBA), while potassium persulfate (KPS) was employed as the polymerization initiator. The bleached pulps, DACs, and superabsorbent composites (SAPs) hydrogels were

characterized by FTIR, XRD, SEM, and TGA. The results showed that an increase in the periodate equivalent relative to the glucose residue resulted in an increase in aldehyde content and a decrease in crystallinity, accompanied by a change in the morphological properties of the obtained oxidized fibers. Moreover, SAPs containing dialdehyde cellulose exhibited greater porosity compared to SAPs without DACs. The swelling capacity of the hydrogels was determined using the tea bag method. The SAPs of DAC extracted from banana fibers demonstrated a higher water absorption capacity of 1300 g g⁻¹ compared to other SAPs due to the specific hydrophilic properties of the DAC derived from banana, especially its lower crystalline index.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10570-025-06652-2>.

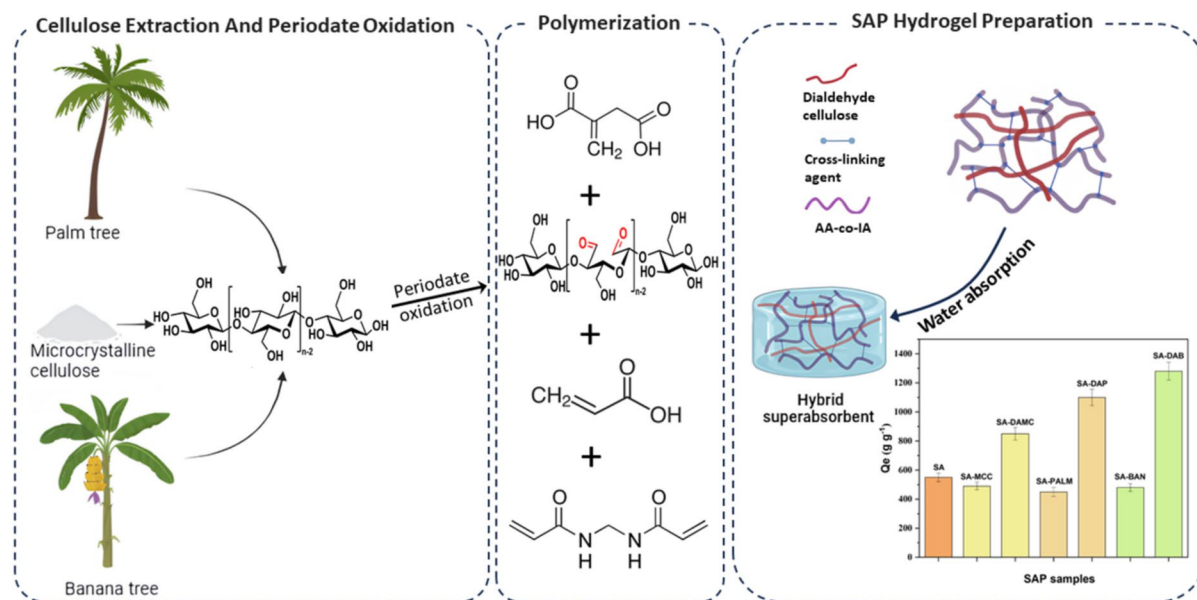
Y. Chaab · H. Aitbella · A. Y. Boussif · M. Bezbiz ·
M. Latifi · L. Belachemi · H. Kaddami (✉)
Cadi Ayyad University, FST – Gueliz, Innovative Materials
for Energy and Sustainable Development (IMED-Lab),
549, Avenue Abdelkrim Elkhatabi, Marrakech 40000,
Morocco
e-mail: h.kaddami@uca.ma

H. Ben Youcef
Mohammed VI Polytechnic University, HTMR,
Ben Guerir 43150, Morocco

H. Kaddami
Mohammed VI Polytechnic University, SUSMAT-RC,
Ben Guerir 43150, Morocco

A. Y. Boussif · M. Bezbiz · C. Moreau · B. Cathala
UR1268 BIA, INRAE, Nantes 44316, France

Graphical Abstract



Keywords Cellulose source · Dialdehyde cellulose · Superabsorbent composites · Water swelling · Comparative study

Introduction

Agricultural activities generate vast amounts of crop waste each year, posing a significant environmental challenge. Among these, banana and palm waste are particularly abundant due to the high global demand for these crops (Putra et al. 2022). For instance, banana cultivation results in significant by-products, such as pseudostems and leaves, while palm cultivation generates fronds and trunks as waste. These materials are often underutilized or improperly disposed of, leading to environmental pollution and wasted potential (Sayyed et al. 2021). To address this issue, valorizing such crop waste has become a critical area of research, focusing on extracting valuable components like cellulose (Castro et al. 2024; Sharma et al. 2024).

Cellulose and its derivatives have been extensively investigated, and promising potential for developing environmentally friendly SAP materials has been demonstrated (Chavan et al. 2024). Banana fibers contain about 60–85% in their cell walls, along with a crystallinity index of 47–65% (Akatwijuka et al. 2024). On the other hand, palm fibers have a cellulose content of around 29–40% and demonstrate a higher CI of 67% (Elseify et al. 2019). The variation in cellulose content and crystallinity within plants and the diversity of parameters influencing its extraction highlight the diversity of characteristics of cellulose obtained from various plant species and the diversity of applications. Chemical treatments can create microcrystalline cellulose, which enhances the crystallinity index of cellulose obtained from different sources by transforming it into a refined form with a compact crystalline structure. This can increase the cellulose's crystallinity to 77–95%, thereby improving its overall crystallinity. These treatments efficiently eliminate disordered areas, enhancing the cellulose crystallinity and stiffness (Hamdan et al. 2019; Lupidi et al. 2023). Understanding these variations is crucial for informing the development of sustainable

materials and the design of products. Cellulose generated from bananas and palms has been utilized in biodegradable polymers research, food packaging, biofuel research, textiles, multilayered papers, and agronomy (Raza et al. 2022; Saud et al. 2023; Kumar et al. 2023).

In modern agriculture, cellulose has gained attention as a core component of SAPs for water management (Yang et al. 2024). SAPs are hydrogels capable of absorbing and retaining water hundreds of times their own weight, releasing it gradually to plants as needed (Oladosu et al. 2022; Saraswathy et al. 2023). This technique, particularly promising for sustainable irrigation, has become a focal point in addressing water scarcity issues in agriculture (Behera and Mahanwar 2020; Takahashi et al. 2023). By using cellulose extracted from crop waste, such as banana and palm residues, it is possible to create hybrid SAPs that combine sustainability with high water absorption performance and biodegradability. This approach contrasts with synthetic polymers like polyacrylic-based SAPs, derived from fossil fuels such as acrylic acid (AA) and its derivatives, which dominate the market despite their low degradability and lack of sustainability.

Cellulose's hydrophilic nature can be further enhanced through chemical modifications, such as oxidation. Dialdehyde cellulose (DAC) is a notable derivative synthesized through selective oxidation using sodium periodate. DAC incorporates aldehyde groups, facilitating cross-linking in polymer networks and significantly improving water absorption capacity (Simon et al. 2022). Similarly, microcrystalline cellulose (MCC), derived through chemical treatment of cellulose, exhibits high crystallinity, stiffness, and compatibility for use in SAP formulations. Integrating DAC into SAPs can develop hybrid materials with enhanced biodegradability, biocompatibility, and water absorption properties (Pérez-Álvarez et al. 2018).

Hybrid SAPs, which combine synthetic polymers like polyacrylic acid PAA with natural polymers such as polysaccharides and naturally resourced components, have shown remarkable performance improvements. Such advancements make them highly attractive for agricultural applications, acting as efficient water reservoirs, reducing irrigation frequency, and conserving water resources. In addition, hybrid SAPs possess excellent thermal stability, environmental

responsiveness, and reusability, making them suitable for diverse applications beyond agriculture, including horticulture and controlled drug delivery (Călina et al. 2021).

Cellulose-based hybrid SAP materials can absorb up to 560 g g^{-1} of water in distilled conditions, with swelling degrees reaching 6000% within hours (Zhang et al. 2023). Wang et al. reported an absorbance capacity of 750 g g^{-1} for SAPs made from acrylic acid, carboxymethyl cellulose and graphene oxide (CMC-g-PAA/GO) (Wang et al. 2017). Moreover, Hydroxyethylcellulose copolymer with acrylate acrylamide (P(AA-co-AM)) were reported to absorb up to 240 g g^{-1} in sandy soil (Song et al. 2019).

Other research has focused on the copolymerization of AA with itaconic acid (IA), a naturally derived polymer synthesized through the fermentation of carbohydrates. Due to its trifunctional structure, IA facilitates the creation of various SAPs with enhanced binding properties. In addition to improving the performance of SAPs, IA exhibits several beneficial biological activities, such as antimicrobial, anti-inflammatory, and antitumor effects. Furthermore, IA retains key advantages like environmental friendliness, biocompatibility, and sustainability, making it a highly attractive material for a range of applications, including agriculture, healthcare, and environmental management (Kwon et al. 2021; Teleky and Vodnar 2021).

However, most studies on the copolymerization of acrylic and IA have reported relatively low absorbency. For instance, when IA, AA, and 2-hydroxyethyl methacrylate were copolymerized, the resulting material demonstrated a methylene blue dye removal capacity ranging between $197\text{--}260 \text{ mg g}^{-1}$ (Oyarce et al. 2020). A significant study recently explored the copolymerization of polyacrylic acid, IA, and cellulose derivatives (DAC). This work produced SAPs with an impressive water absorbency of 1240 g g^{-1} , which is a remarkable result (Boussif et al. 2024). However, the previous study used only a single cellulose source, prompting our investigation into the performance of hybrid SAPs derived from different cellulose sources.

This study aims to identify the cellulose type that best supports the targeted application, offering insights into optimizing material selection for high-performing, sustainable SAPs. Specifically, we will explore the potential of cellulose materials from banana and palm waste, alongside commercial MCC, in developing

polyacrylic acid-based hybrid SAPs. The performance of these SAPs will be evaluated by comparing their swelling properties, water absorption, and overall sustainability.

Experimental methods

Materials

Cellulose was extracted from date palm and banana tree fibers, and microcrystalline cellulose (MCC) powder ($M_w = 220,000 \text{ g mol}^{-1}$, $DP = 1350$) was purchased from Sigma Aldrich. Sodium chlorite (NaClO_2), Sodium hydroxide (NaOH , 99%), Sodium periodate (NaIO_4 , 99.8%), hydroxylamine hydrochloride ($\text{HONH}_2\cdot\text{HCl}$), ethylene glycol, acrylic acid (AA, 99%), potassium persulfate (KPS: $\text{K}_2\text{S}_2\text{O}_8$, 99%), and *N,N*-methylene-bis acrylamide (MBA), were supplied by Sigma Aldrich. Acetone, Acetic acid, and sulfuric acid were purchased from VWR CHIMICALS. Itaconic acid (IA, 99%) was purchased from Thermo Fisher Scientific.

Preparation of cellulose pulp from date palm and banana

The cellulose pulp was prepared using the methodology outlined in the published literature (Bendahou et al. 2009; Benhamou et al. 2018; El Idrissi et al. 2022; Aitbella et al. 2024).

The plant material derived from the date palm leaflet was subjected to coarse grinding using an electric grinder (350 g, DAMAI, China) and a drying process (70 °C, 12h). The fibers were Soxhlet-extracted for 24 h with 200 ml of acetone to remove pigments, lipids, waxes, and all other substances soluble in organic solvents. Subsequently, the decolorized fibers were air-dried. The banana pseudo-stem was extracted using a decortication machine, which combined water retting and scraping. The initial step involved the separation of fiber bundles from residual components. The fibers were then thoroughly water-washed and dried.

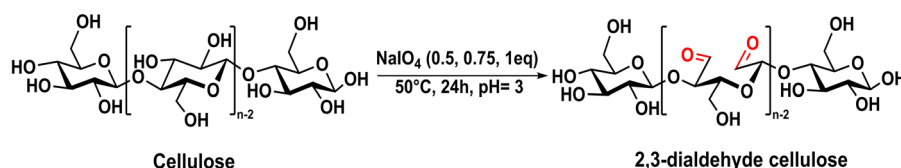
Both types of fibers then underwent the same separation process. They were sieved with a 400- μm sieve and then subjected to a chemical treatment with sodium hydroxide (NaOH) at a concentration of 3% (w/v), performed three times at 70 °C. Subsequently, the fibers underwent bleaching with a solution comprising sodium chlorite (NaClO_2), and an acetic buffer at a pH level of 4.8. The mixture was heated to 70 °C for two hours, allowing the elimination of lignin due to the oxidizing activity of sodium chlorite. At the end of the process, the suspension was filtered and thoroughly washed with water. The bleaching process was repeated until the cellulose fibers appeared white. The removal of lignin was verified using a sulfuric acid test following the Klason lignin method (Bendahou et al. 2007; Sbiai et al. 2011). The bleached pulps were obtained after three cycles of bleaching.

Preparation of DAC (2,3-dialdehyde cellulose) and yield calculation

The cellulose dialdehydes were prepared by adding 1g of cellulose extracted from date palm, banana, and MCC (corresponding to 6.21 mmol of glucose residue of cellulose) to 50 ml of distilled water. This mixture was sonicated for 30 min in an ultrasonic bath, and the pH was adjusted to 3 using 0.1 M acetic acid. Oxidation was carried out with different molar ratios NaIO_4 / glucose residue (0.5, 0.75, and 1 eq). First, sodium periodate was added to the sonicated cellulose in complete darkness. The reaction was conducted in the dark for 24 h under magnetic stirring at 50 °C (Fig. 1), following the methodology previously reported by our research group (Bezbiz et al. 2024; Boussif et al. 2024; Aitbella et al. 2024).

At the end of the reaction, 1 eq of ethylene glycol was added to eliminate the excess residual periodate, and stirring was continued for 1 h at room temperature. Finally, the obtained DAC was separated by centrifugation at 10,000 rpm and washed several times with distilled water until the pH of the washing water was neutral. The final materials (DACs) obtained from

Fig. 1 Reaction scheme of the regioselective oxidation of cellulose by sodium periodate



bleached date palm pulp, banana pulp, and MCC were designated by DAP, DAB, and DAMC, respectively.

The reaction yields during oxidation were calculated using Eq. (1).

$$\text{Yield (\%)} = \frac{W_f}{W_i} \times 100 \quad (1)$$

W_f : the final weight of DAC, determined by weighing the product after drying until complete water evaporation.

W_i : the initial weight, representing the weight of the raw material (bleached pulp) introduced into the reaction.

SAPs hydrogels synthesis

The synthesis of SAPs was conducted using chain radical polymerization in an aqueous medium following the conditions optimized in our previous studies (Bezbiz et al. 2024; Boussif et al. 2024). The primary process involved copolymerizing AA with IA in the presence of bleached pulp or DAC derived from different cellulose sources to permit the grafting of the polymer chains on their surfaces. It's important to note that the DACs used for the SAPs preparation are the DACs obtained with a molar ratio $\text{NaIO}_4/\text{glucose residue}$ of 0.5. The polymerization took place in the presence of potassium persulfate (KPS) as the initiator and N, N-methylene-bis-acrylamide (MBA) as the crosslinker (Bora and Karak 2022). Firstly, 30 wt.% sodium hydroxide (NaOH) was used to modify the ratio of carboxylic acid groups to carboxylate groups in AA. Subsequently, in a 100 ml three-neck round-bottom flask with a mechanical stirrer and under a nitrogen environment, AA and IA were added to 10 ml of distilled water. The solution was stirred constantly as it heated to 85 °C. A 5wt.%

of BP/DAC suspension from various sources was then introduced to the mixture. Then, the crosslinker MBA and the initiator KPS were sequentially added, and the polymerization reaction was performed by stirring the resulting mixture for one additional hour at 85 °C.

Upon completion, the hydrogels were washed with pure ethanol and dried at 50 °C until a stable mass was achieved, making them ready for further studies and tests. Sample codes are provided in Table 1.

Chemical composition of date palm and banana fiber

In this section, we outline the experimental protocols used to analyze the chemical composition of banana and date palm fibers. The study was conducted following French Standards (NFT12-011) (AFNOR 1979). The lignin content was determined using the Klason method (Supplementary Information S1), while the weight fractions of cellulose, hemicellulose, and lignin were measured in triplicate to ensure accuracy and reproducibility (Bendahou et al. 2007; Sbiai et al. 2011).

Aldehyde group titration

The oxidation by periodate was characterized by a degree of oxidation (DO), which represents the number of aldehyde functions introduced per glucose residue and the percentage of oxidation, indicating the ratio of oxidized units over the total number of glucose residue. The maximum DO that can be achieved is 2, corresponding to a fully oxidized DAC (100%), which equates to approximately 12.5 mmol of aldehyde groups per gram of cellulose. The aldehyde content was determined using the oximation method

Table 1 Formulation, weight fractions (wt%), and codes of the SAPs

Sample name	AA/MBA	IA	BP	Dialdehydes cellulose 0.5 eq
SA	85 wt% (84.5wt%/0.5wt%)	15 wt%	–	–
SA-MCC	80 wt% (79.5wt%/0.5wt%)	15 wt%	5 wt%	–
SA-Palm	80 wt% (79.5wt%/0.5wt%)	15 wt%	5 wt%	–
SA-Ban	80 wt% (79.5wt%/0.5wt%)	15 wt%	5 wt%	–
SA-DAMC	80 wt% (79.5wt%/0.5wt%)	15 wt%	–	5 wt%
SA-DAP	80 wt% (79.5wt%/0.5wt%)	15 wt%	–	5 wt%
SA-DAB	80 wt% (79.5wt%/0.5wt%)	15 wt%	–	5 wt%

(Kim et al. 2000; Dang et al. 2019; Bezbiz et al. 2024; Boussif et al. 2024).

In brief, 0.1 g of oxidized cellulose (DAC) was combined with 25 mL of a hydroxylamine hydrochloride solution (0.2 M). The mixture was subjected to sonication for 30 min, then stirred and heated at 40 °C for 18 h. The reaction mixture was subsequently transferred to an ice-water bath to stop the reaction. Later on, the HCl formed was titrated with a sodium hydroxide solution (NaOH, 0.2 M). A blank measurement was performed similarly with 0.1 g of unoxidized cellulose.

The aldehyde content was determined using the following Eqs. (2):

$$\text{AC}(\text{mmol/g}) = \frac{C_{\text{NaOH}} \times (V_{\text{DAC}} - V_{\text{C}})}{m} \quad (2)$$

where: m is the weight of DAC, C_{NaOH} is the Concentration of the NaOH solution, V_{DAC} is the equivalence volume of oxidized cellulose, V_{C} is the equivalence volume of non-oxidized cellulose.

Measurement of water absorption capacity of SAPs

The maximum water swelling capacity (Q_m) of the synthesized SAPs was ascertained using the tea bag method (Lu et al. 2015). A predetermined amount of dry SAPs was weighed (m_0) and placed inside a tea bag. After immersing the tea bag in distilled water at 25 °C for a time (t) until swelling equilibrium was attained, the swollen sample was weighed (m_t). An unfilled tea bag was used as a control in the experiment. The water absorption capacity at equilibrium (Q_{eq}) of the SAPs was calculated using Eq. (3), which considers the weight difference before and after submersion in water:

$$Q_{\text{eq}}(\text{g/g}) = (m_t - m_0)/m_0 \quad (3)$$

where m_0 and m_t are the weights of the dry and swollen samples, respectively.

Fourier transform infrared (FT-IR) spectroscopy.

FTIR spectra were obtained at room temperature using a PerkinElmer Spectrum 1000 FTIR spectrometer, with continuous scan mode under dry air. A KBr pellet was prepared as the background

spectrum, and pellets containing the samples were prepared in KBr matrices (2 mg of ground product in 198 mg of KBr). FTIR spectra were recorded with 32 scans within the 450–4000 cm^{-1} range at a resolution of 4 cm^{-1} .

X-ray diffraction (XRD)

X-ray diffraction (XRD) characterization was performed with a Rigaku model SmartLab X-ray diffractometer to identify the crystal structure of the samples within a 2θ range from 5 to 40°. The crystallinity index (CrI) was calculated from the height of the 200 peak (I_{200} , $2\theta=22^\circ$) and the intensity minimum between the 200 and 110 peaks (I_{am} , $2\theta=18^\circ$) using the empirical Eq. (4) of Segal et al. (1959)

$$\text{CrI}(\%) = \frac{I_{200} - I_{\text{am}}}{I_{200}} \times 100 \quad (4)$$

Thermogravimetric analysis (TGA)

Thermogravimetric analysis (TGA) was performed to determine the thermal stability of the polymeric materials using a TA Instruments Q500 analyzer. The study was conducted over a temperature range from 25 to 600 °C with a heating rate of 10 °C.min⁻¹ under a nitrogen atmosphere.

Scanning Electron Microscopy (SEM)

The morphologies of BPs, MCC, DACs, and their SAPs were examined using a Vega 3 TESCAN scanning electron microscope equipped with a conventional tungsten heated cathode, suitable for both high vacuum and low vacuum operations. All samples were sputter-coated with carbon in argon to enhance their conductivity and were examined in HI-VAC mode with a probe current of 6 mA at a 10 kV acceleration.

Results and discussion

Chemical composition of date palm and banana fibers

The chemical composition of lignocellulosic fibers from date palm and banana fibers is summarized

Table 2 The chemical composition of date palm and banana fibers (% weight)

Chemical composition	Average rates (%) [*]	
	Palm date	Banana fibers
Cellulose	39 ± 2	52 ± 2
Lignin	26 ± 2	21 ± 3
Hemicellulose	25 ± 2	19 ± 2
Ash	7 ± 1	4 ± 1
Others ^{**}	ND	ND

^{*}The average rates were calculated after triplicate. Taking into account the humidity content of the fibers

^{**}Pectin, water-soluble sugars

in Table 2. These results represent averages from at least three trials for each element. The data confirm that both date palm and banana fiber are primarily composed of three main components with different proportions: cellulose, hemicelluloses, and lignin. Additionally, extractable compounds are also present in their composition. Cellulose is the predominant constituent, accounting for approximately 39% of date palm fibers and 52% of banana fibers. Lignin and hemicelluloses account for around 21–26% and 19–25% of the fiber composition, respectively. The present findings are in accordance with those previously established by other researchers. For example, Sbiai et al. (2011) reported almost similar cellulose, hemicellulose, and lignin wt% of 35, 28, and 27%, respectively, in date palm fibers. However, the pseudo banana fiber has a high cellulose content of 56.48%, as mentioned by Amutha et al. (2022), which is closely aligned with the 52% cellulose found in banana fibers in this study. The cellulose content in

date palms is lower than in banana fibers, as shown in Table 2. However, it can be noted that the hemicellulose and lignin content in banana fibers is lower compared to those in date palms. Akatwijuka et al. (2024) have reported a similar observation concerning the chemical composition of date palm and banana fibers. In addition, other researchers partially agree with these results (Elseify et al. 2023). Lai et al. (2023), who examined the thermal properties of banana fibers, mentioned that cellulose content constituted 60–65%, while the hemicellulose and lignin content was 6–19% and 5–10%, respectively. Moreover, Beroual et al. (2021) reported the lowest cellulose content in date palm at 31.4% compared to this study, while the hemicellulose and lignin contents were 20.9% and 25.2%, respectively. The published studies showed different results for both lignocellulosic fibers in terms of their chemical composition, including cellulose, hemicellulose, and lignin, mainly due to differences in soil and climatic conditions in which the biomass was grown (Alotaibi et al. 2019).

DAC preparation and characterization

Aldehyde content and yield loss: effect of periodate concentration

The aldehyde content in DAC and the yields of reactions were assessed to evaluate the efficiency of periodate regioselective oxidation of cellulose. Table 3 presents the properties of DAMC, DAP, and DAB, produced under different periodate equivalent moles numbers at 50 °C and pH = 3 for 24h.

The impact of periodate on the chemical structure of cellulose is significant. As the periodate

Table 3 Results of the yield and aldehyde titration of DACs of microcrystalline cellulose, banana pulp, and date palm pulp at different periodate concentrations

Raw materials	NaIO ₄ equivalent	Aldehyde Contents (mmol/g)	Yield (%)	Degradation (%)
MCC	0.5	2.5 ± 0.1	63.7 ± 0.2	36.3 ± 0.2
	0.75	3.6 ± 0.5	44.9 ± 0.2	55.1 ± 0.2
	1	5.3 ± 0.2	24.9 ± 0.2	75.1 ± 0.2
Bleached pulp of Date palm	0.5	2.8 ± 0.1	64.0 ± 0.2	36.0 ± 0.2
	0.75	4.4 ± 0.2	61.7 ± 0.2	38.3 ± 0.2
	1	6.3 ± 0.5	54.5 ± 0.2	45.5 ± 0.2
Bleached pulp of Banana	0.5	3.3 ± 0.5	66.5 ± 0.1	33.5 ± 0.1
	0.75	5.7 ± 0.4	59.9 ± 0.1	40.1 ± 0.1
	1	9.5 ± 0.3	57.3 ± 0.7	42.7 ± 0.7

equivalent increased from 0.5 to 1.0, the aldehyde content of DAC increased in proportion. Conversely, the weight lost after the reaction increased significantly with increasing periodate equivalents for the three cellulose types. This weight loss is attributed to the degradation of cellulose chains and the breaking of C–O–C bonds linking the monomers, resulting in the formation of water-soluble compounds under strong oxidizing conditions (Siller et al. 2015). These results are consistent with the work of Errokh et al. (2018), who found that increasing oxidation intensity resulted in a significant decrease in yield due to degradation of the cellulose structure. Moreover, the relationship between oxidation conditions and cellulose yield observed in this study is consistent with the conclusions of Sun and Jiang (2024), who found that stronger oxidation conditions result in reduced yield and increased aldehyde content due to more aggressive cleavage of glycosidic bonds. As periodate oxidation intensifies, it specifically targets the boundaries between amorphous and crystalline regions, producing shorter cellulose chains (Chen and van de Ven 2016). Notably, the two bleached cellulose pulps exhibited comparable outcomes concerning degradation when the periodate concentration was increased.

Compared to MCC, they demonstrated greater resilience to oxidation. While this phenomenon may partly be attributed to the molecular weight and purification process of MCC, which often involves acid hydrolysis (Ventura-Cruz et al. 2020), it is evident that other structural factors also play an essential role. The average molecular weight which generally decreases with an increasing the periodate equivalent (Nypelö et al. 2021), which can be attributed to the depolymerization of DACs by side reactions (Hosoya et al. 2018; Sun and Jiang 2024). The fiber dimensions of bleached pulps in comparison to the MCC powder may also contribute to their greater resilience, as it includes fractions that are less vulnerable to oxidative cleavage. This is illustrated in the recent work of Ghodake et al. (2023) which also demonstrates that MCC has a high specific surface area and a lower polydispersity index PDI compared to cellulose fibers.

FTIR analysis

As can be observed in Fig. 2 (a, b, and c), all three oxidized cellulose samples show similar FTIR spectral bands. The band at 3413 cm^{-1} is attributed to the

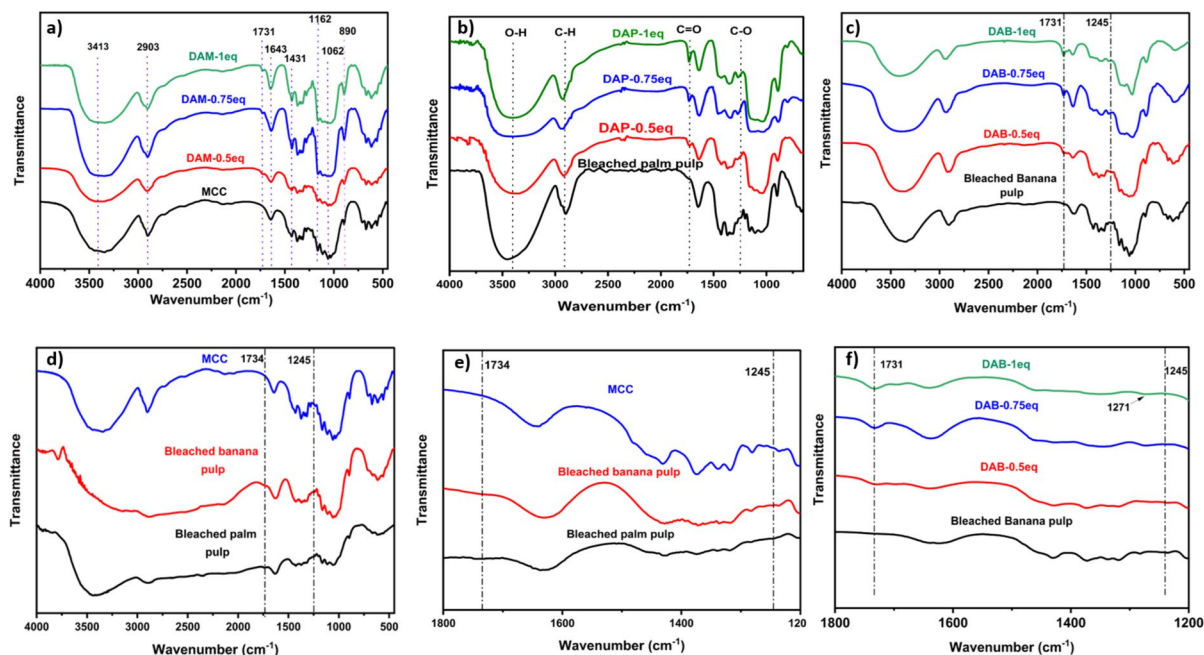


Fig. 2 FT-IR spectra of the oxidized **a** microcrystalline cellulose **b** bleached palm pulp **c, f** bleached banana pulp by different periodate concentrations, and **d, e** unoxidized microcrystalline cellulose, bleached banana, and palm pulps

stretching vibration of hydroxyl (O–H) groups, while the band at 2903 cm^{-1} corresponds to the vibrational bands of C–H in CH_2 and CH groups. The bending vibration band at 1643 cm^{-1} represents the interaction between cellulose and adsorbed water molecules. The bands at 1431 cm^{-1} and 1062 cm^{-1} are attributed to the deformation frequencies of the $-\text{CH}_2$ group and C–O stretching vibration, respectively (El-Sakhawy et al. 2018). The conversion of cellulose to Dialdehyde cellulose could be confirmed by the presence of the aldehyde carbonyl vibration band at 1731 cm^{-1} in all FTIR spectra of DACs (0.5, 0.75, and 1 eq) (Azzam et al. 2015), the intensity of this band increases with the aldehyde content in the oxidized cellulose. The band at 890 cm^{-1} corresponds to the absorption of hemiacetal groups formed between the aldehyde and alcohol functions. Subsequently, the bleached banana and palm pulp spectra exhibit a minor band at 1734 cm^{-1} , which can be attributed to the C=O stretching of the acetyl group and uronic ester groups of hemicellulose or to the ester linkage of the carboxylic group in the ferulic and p-coumaric acids of lignin and/or hemicellulose. Meanwhile, the band at 1245 cm^{-1} is attributed to the C–O stretching of the aryl group in lignin (Le Troedec et al. 2008). The aforementioned bands are absent in the MCC spectrum and are more intense in the bleached palm pulp than in banana pulp, as illustrated in Fig. 2 (d, e). Furthermore, Fig. 2f depicts the disappearance of these two bands in all DACs spectra and the emergence of a new band at 1271 cm^{-1} , which was assigned to the C–O of hemiacetal groups. These findings indicate that the bleaching process is unable to

completely remove residual hemicellulose and lignin residual from the surface of cellulose. However, the majority of hemicellulose and lignin were successfully removed from MCC and DACs through acid hydrolysis and periodate oxidation treatments, respectively (Zuluaga et al. 2009; Benhamou et al. 2015).

X-ray diffraction XRD

XRD was used to characterize the variation in the crystallinity of the cellulose after the periodate oxidation process. Figure 3 illustrates the XRD patterns of all cellulose types, including their dialdehyde cellulose counterparts. The MCC patterns display distinct peaks at $2\theta \sim 15^\circ$, 16.4° , 22.5° and 34.5° and correspond to the (1 $\bar{1}$ 0), (110), (200), and (004) crystallographic planes of cellulose I. However, the presence of the peaks at $2\theta \sim 13^\circ$ and 20.2° in the MCC pattern can also suggest that the commercial MCC used is a mixture of Cellulose I and Cellulose II (French 2014; Sun et al. 2016; Huang et al. 2020). In contrast, the patterns of date palm and banana pulps exhibit a broad peak at 15.7° and 15.6° , respectively. This is probably due to the superimposed peaks at $2\theta \sim 15^\circ$ and 16.5° . This suggests that MCC has greater crystallinity compared to the pulps, resulting in more distinct and sharper XRD peaks.

For all cellulose types, the intensity of the peaks at 2θ of 22° decreases while comparing the spectra cellulose to DACs, this intensity further decreases when periodate equivalent used for the oxidation increases. This indicates a successive cleavage of the $\text{C}_2\text{--C}_3$ pyranose rings in cellulose, leading to the formation

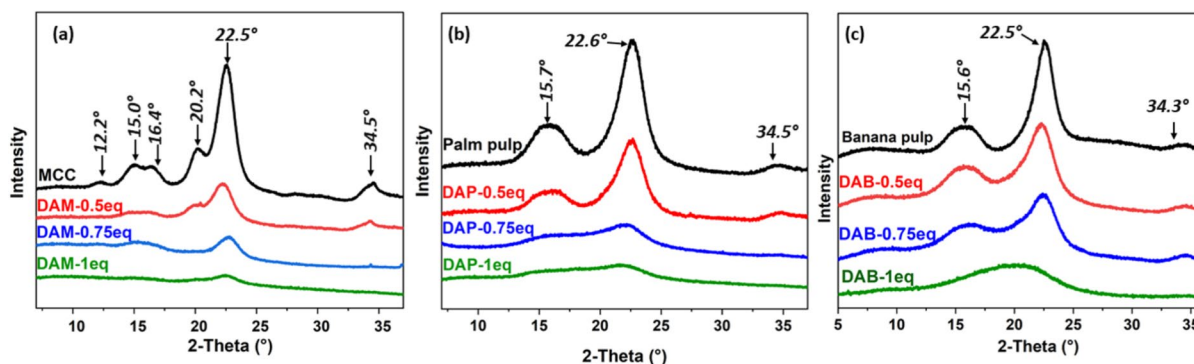


Fig. 3 X-ray diffraction (XRD) patterns of the cellulose oxidation of **a** MCC, **b** bleached palm pulp, and **c** bleached banana pulp with varying periodate concentrations

Table 4 the crystallinity index of the obtained dialdehyde cellulose from palm, banana, and microcrystalline cellulose as a function of the periodate/glucose residue ratio

Cellulose sources	Moles equivalent			
	0 eq	0.5 eq	0.75 eq	1 eq
<i>Palm date pulp</i>	69%	48%	27%	12%
<i>Banana pulp</i>	63%	41%	22%	ND
<i>Microcrystalline cellulose</i>	78%	51%	40%	18%

of more free amorphous regions at the ends of cellulose chains (Yu et al. 2023). This results in a decrease in cellulose crystallinity, as shown in Table 4. Moreover, the peak at approximately 22° exhibits a more pronounced decline in the XRD patterns of DAC 1eq, indicating that these samples have completely transitioned to an amorphous state (Ruan et al. 2016). This trend is normal and could be explained by the increase of the cellulose chain's mobility and flexibility after degradation in the amorphous domains. This increased mobility disturbs the adjacent crystal stability and leads to its breaking down.

Scanning electron microscopy SEM

Figure 4 depicts the SEM images of the DACs samples of different degrees of oxidation prepared by different periodate/glucose residue ratios. It can be observed that the morphological alterations induced by periodate oxidation were more pronounced when the oxidation degree increased. For example, the initial date palm samples exhibited microfibril structures with widths of a few micrometers (around 10 µm) (Fig. 4b-1). After the oxidation (Fig. 4b-2), the diameter of the fibers is significantly decreased to less than 5 µm. At a higher degree of oxidation, the diameter becomes lower than 3 µm, and a non-fibrous structure becomes more and more visible with the increase of the oxidation degree. This non-fibrous is probably the result of the degradation and the increase of the amorphousness of the sample, as shown by XRD analyses. The deterioration of the fibrous structure when the oxidation degree is increased is more visible for the MCC sample than for the other samples. Similar observations could be made in the SEM images of banana fiber with different oxidation degrees. These observations align with the observations previously reported from the XRD analyses.

The pore size between the fibers may also be affected following cellulose treatment (Sayyed et al. 2018). In this instance, it was observed that an increase in the periodate content led to a decrease in the pore size between the fibers. Furthermore, the DAC fibers reveal a smooth surface, in contrast to that of virgin fibers, which display a rough texture. demonstrate the shrinkage and compaction of the fiber surface during oxidation, which is consistent with the loss of crystalline domains.

SAPs preparation and characterization

FTIR analysis

FTIR analyses were also conducted to study the structure of the synthesized SAP hydrogels. Figure 5 shows the FTIR spectrum of different SAPs. The absorption band at 3449 cm⁻¹ is attributed to the stretching vibrations of hydroxyl groups. The band at 2949 cm⁻¹ is assigned to symmetric and asymmetric stretching vibrations of CH₂ groups. The band observed at 1720 cm⁻¹ is attributed to the C=O stretching vibration of carboxylic acid groups; this band overlaps with the NH vibration band of the crosslinking agent and the aldehyde carbonyl vibration band at 1731 cm⁻¹ for the DACs-based SAPs spectra. Additionally, the band at 1630 and 1560 cm⁻¹ is attributed to the surface adsorbed water, and the carboxylate groups, respectively.

Scanning electron microscopy (SEM)

The morphology of the synthesized SAPs was analyzed using scanning electron microscopy SEM. Figure 6 displays SEM images of SAPs derived from bleached pulps and DAC. The microstructure of the SAPs reveals closed pores of varying sizes and distinct surface relief features, with dispersed cellulose fibrils present on the surface.

SEM images show the presence of cellulose fibrils inside and on the surface of the hydrogel (indicated with white arrows in images), highlighting its porous structure. However, the SAPs derived from DAC exhibit a more homogeneous structure compared to those based on bleached pulps, attributed to the oxidation process, which reduces the diameter of cellulose microfibrils and improves their dispersion in the poly (Acrylic co-Itaconic) acid polymer. This

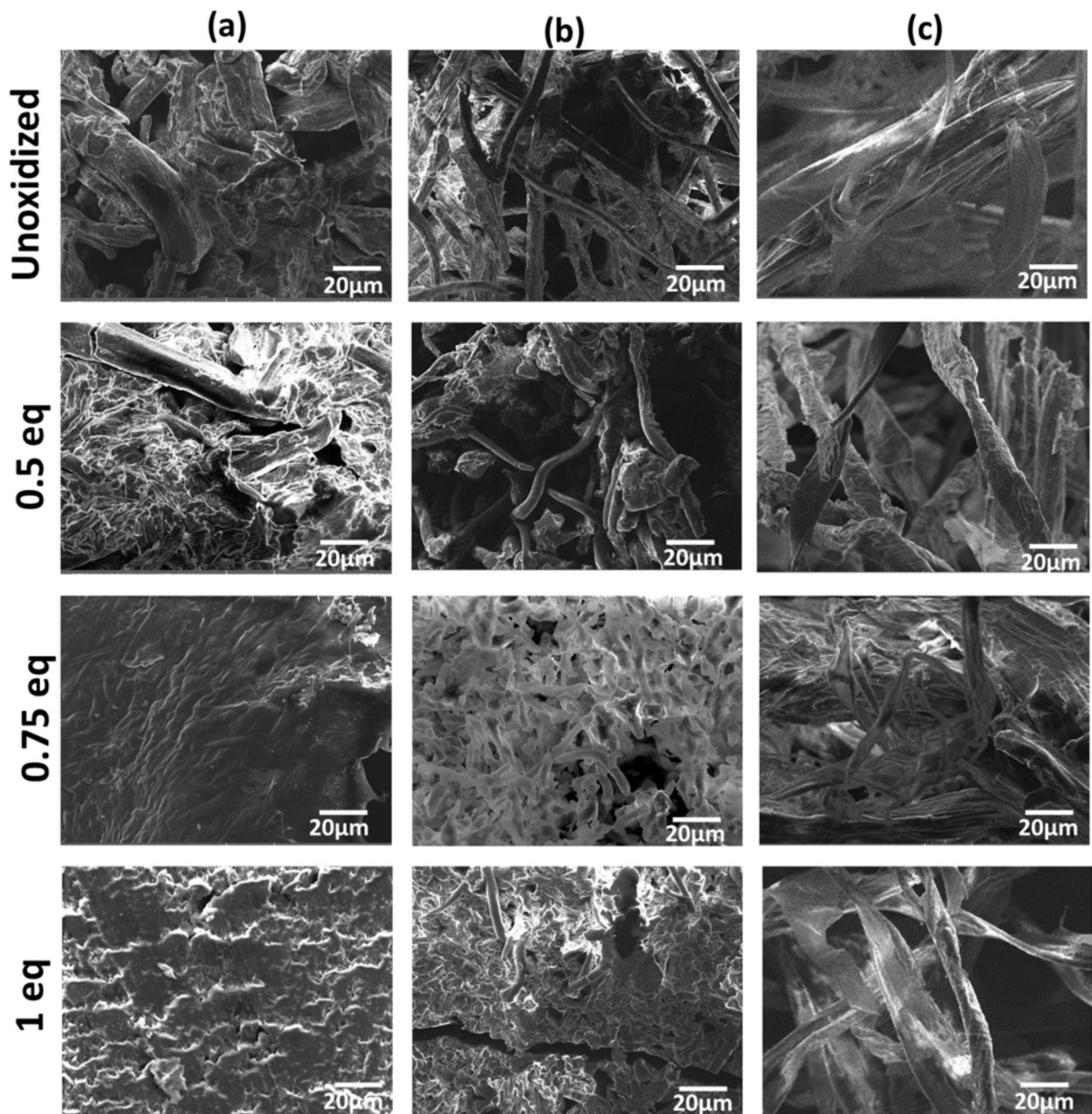


Fig. 4 SEM images **a** MCC **b** Bleached palm date pulp and **c** Bleached banana pulp, oxidized with different periodate concentrations

homogeneous structure of DAC based SAP is propitious to swelling properties improvement in comparison to cellulosic fibers-based SAPs. On the other hand, while comparing the different SAP, it seems that the pore size (cycles in yellow) increases when comparing SAP of DAB to the two other SAPs. These

changes are crucial for water absorption, retention, and swelling properties.

Thermogravimetric analysis (TGA)

Figure 7 presents the thermogravimetric analysis (TGA) and DTG results for the copolymer of poly

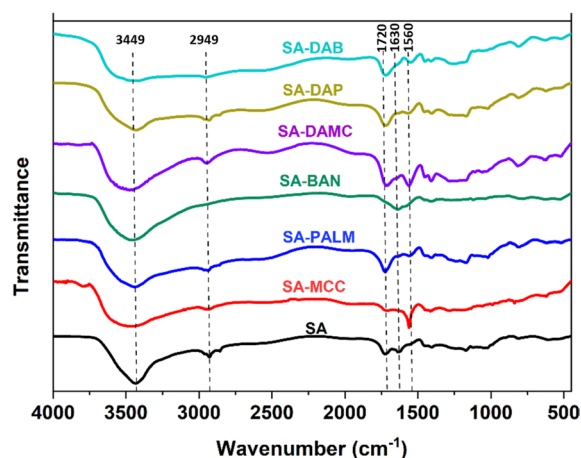


Fig. 5 Infrared spectra of the synthesized superabsorbent polymers SA, SA-MCC, SA-Palm pulp, SA-Banana pulp, SA-DAM, SA-DAP, and SA-DAB

(AA-co IA) and the different hybrid SAP of celluloses (Fig. 7a, c) and DAC (Fig. 7b, d). The examination of the TGA and DTG curves of the copolymer PAA co-IA, performed up to 600 °C, highlights several distinct phases of thermal decomposition.

In fact, below 130 °C, the initial weight loss is attributed to the evaporation of physically adsorbed water. Within the temperature range of 170–350 °C, the SAP (PAA co-IA) undergoes an initial

decomposition of side groups, leading to the release of water and carbon dioxide from the breakdown of carboxyl groups (Kwon et al. 2021). Between 350 and 490 °C, the PAA-IA backbone is significantly degraded, coupled with removing carboxyl groups from the IA units (Chen et al. 2015; Liu et al. 2022). This stage results in the emission of additional volatile compounds, including carbon monoxide, carbon dioxide, and water. Beyond 490 °C, the remaining polymer fragments endure further decompositions, producing char and other high-temperature decomposition products (Dubinsky et al. 2004; Boussif et al. 2024). Notably, the hydrogel (PAA-co-IA) exhibits a residual mass of 35% at 600 °C, indicating a significant presence of stable char and other decomposition products at elevated temperatures.

Both cellulose and SAPs demonstrate similar degradation behaviors. It is difficult to detect significant differences while comparing the degradation profiles. All SAPs, regardless of the source and type of cellulose content, experience an initial weight loss due to moisture evaporation below 130 °C. Followed by the decomposition of copolymer components, which is increased by around 10 °C compared to the SAP (PAA-IA). However, the final stages, occurring above 350 °C, show more similarity in the starting degradation temperature compared to the SAP-PAA-co-IA.

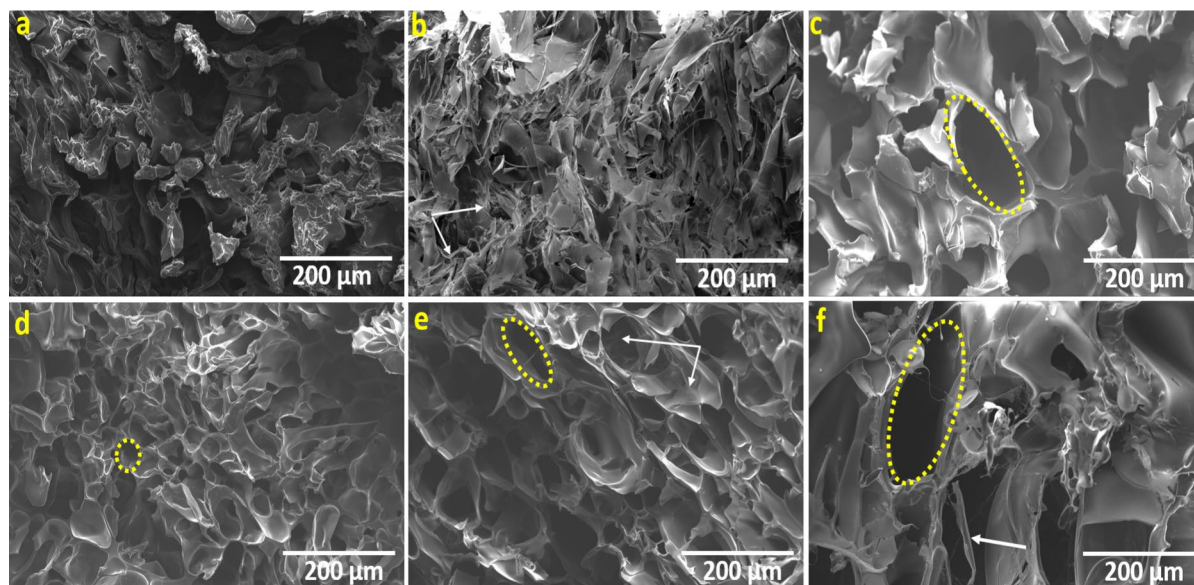


Fig. 6 SEM images of a SAP-MCC, b SAP-Palm, c SAP-Ban, d SAP-DAMC, e SAP-DAP, and f SAP-DAB

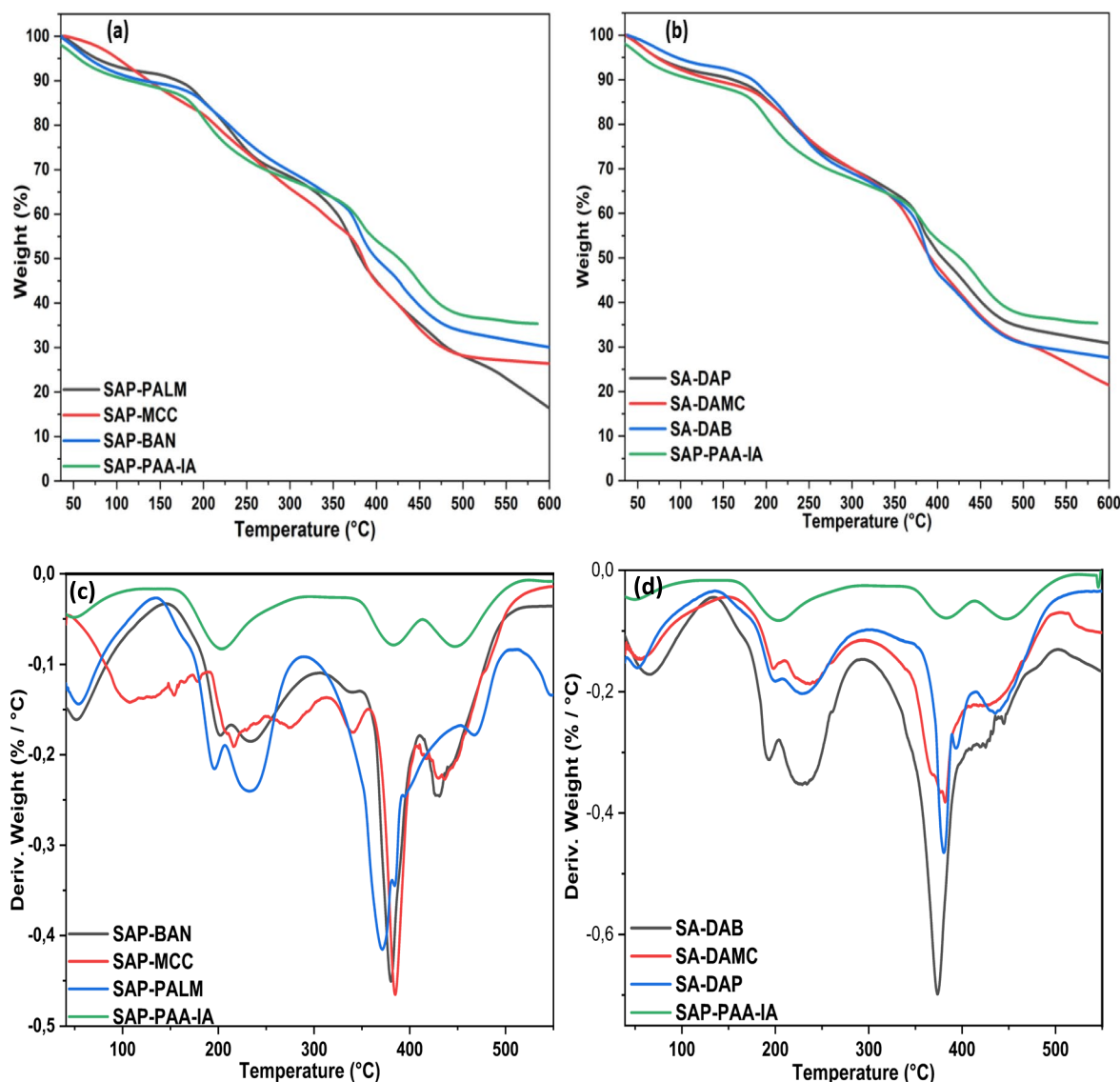


Fig. 7 Thermogravimetric analysis (TGA) of **a** PAA-IA-based superabsorbent incorporating different types of cellulose and **b** PAA-IA-based superabsorbent incorporating different types of DAC

These results suggest that the decomposition pathways of the hydrogel are largely preserved with the addition of cellulose, maintaining the overall thermal decomposition characteristics of the SAPs.

Absorption capacity of SAPs

As described in the “Materials and Methods” session, DACs prepared with a periodate/glucose residue ratio of 0.5 were used to prepare the DAC-based

hybrid SAPs. This choice is justified by the fact that the preparation of these DACs exhibited an optimal balance between yield, crystalline structure, and chemical functionality compared to the other DACs prepared with higher periodate/glucose residue ratio, as presented above. The water absorption experiments were conducted to compare the water absorption properties of the superabsorbent hydrogels prepared based on the cellulose from the different sources with their counterparts based on their oxidized derivatives

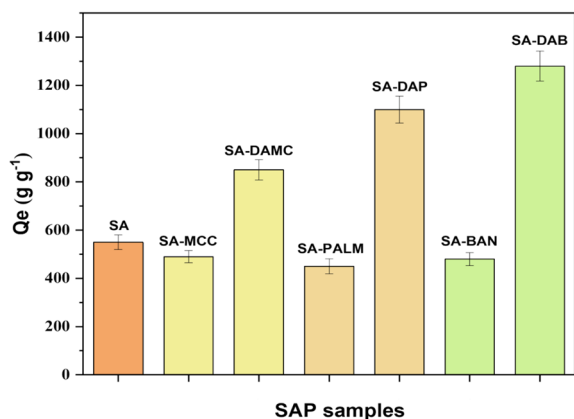


Fig. 8 Equilibrium capacity in distilled water of different synthesized hybrid superabsorbents

DACs. On the other hand, these experiments and comparisons will permit us to highlight and assess how the DACs derived from various cellulose sources impact the characteristics and performances of the resulting SAPs.

Figure 8, depicts the equilibrium capacity of all SAP hydrogels. The figure indicates that the (poly (Acrylic co-Itaconic) acid) copolymer had an absorption capacity of about 550 g g⁻¹. With the introduction of cellulose-bleached pulp, we observed a slight decrease in the absorption capacities of all SAPs. However, the incorporation of dialdehyde celluloses enhanced the absorption capacities for all cellulose sources.

Initially, with bleached pulps, the hybrid SAPs show capacities of about 490, 450, and 480 g g⁻¹ for SA-MCC, SA-PALM, and SA-BAN, respectively. The slight reduction of the absorption values might be due to the insolubility and the limited dispersibility as well as the limited swelling ability of cellulose compared to the Poly (AA-co-IA). This might cause the collapse of the pores of the copolymer, leading to a reduction in the quantities of water molecules penetrating the hydrogel system.

In contrast, cellulose dialdehydes significantly improved the water absorption abilities of the SAPs. The DAC-based hybrid SAPs showed capacities of 850, 1100, and 1300 g g⁻¹ for SA-DAMC, SA-DAP, and SA-DAB, respectively. The DACs were more dispersible, with a lower crystallinity index offering more space for water molecules to penetrate. Additionally, the presence of aldehyde and hemiacetal

groups contributed to more hydrophilic functions (Lv et al. 2024), enhancing the swelling capacities of their SAPs. On the other hand, eliminating hemicellulose and lignin (less hydrophilic than cellulose) after the fibers' oxidation might contribute to the increased absorption capacities of SA-DAP and SA-DAB.

Dialdehyde cellulose (DAC) is obtained through the selective oxidation of cellulose using sodium periodate (NaIO₄). This treatment oxidizes the anhydroglucopyranose units into aldehyde groups, leading to the disruption of intramolecular hydrogen bonds and a reorganization of cellulose chains. This oxidation increases hydrophilicity and water interaction, as aldehyde groups are more hydrophilic than the hydroxyl groups of native cellulose. This enhanced hydrophilicity promotes water absorption and swelling of the material.

DAC plays a key role in improving the water retention performance of superabsorbent hydrogels. This property can be attributed to its distinctive characteristics: an increased pore-forming effect, the hydrophilicity of aldehyde groups, and the ability to interconnect polymer networks through intermolecular acetal and hydrogen bonds (Ruan et al. 2018; Wegrzynowska-Drzymalska et al. 2022; Zou et al. 2023). The pore-forming effect of DAC can be explained by the creation of a three-dimensional porous network during crosslinking. When used in gel formulations with acrylic acid and itaconic acid, DAC can form crosslinked networks through intermolecular acetal and hydrogen bonds (Plappert et al. 2018). This generates a porous structure and facilitates water diffusion, as the hydrophilic functional groups attract water to occupy the available spaces within the polymeric networks. Additionally, DAC interacts with polymer chains formed by acrylic acid and itaconic acid to create crosslinked three-dimensional networks, resulting in good swelling capacity.

These findings are strongly supported by analytical evidence from FTIR, XRD, and SEM. The FTIR spectra revealed an increased intensity of dialdehyde bands, confirming the successful oxidation of cellulose. This chemical modification introduced more hydrophilic functional groups, which is directly correlated with the improved absorption capacity of the hybrid SAPs. XRD analysis provided further insight, showing a decrease in the crystallinity index (CrI) after oxidation. The reduced crystallinity made the cellulose structure less rigid, creating more available

space for water to penetrate, thus increasing absorption. Notably, the CrI of SA-DAB was the lowest among the three samples, followed by SA-DAP and SA-DAMC. This trend aligns with their respective absorption capacities, suggesting that lower crystallinity is a critical factor in achieving higher absorbency.

Moreover, SEM images revealed structural differences that further substantiate these results. The SA-DAB hydrogel exhibited larger pores and more significant spaces for trapping water compared to SA-DAP. Conversely, the smallest pores were observed in SA-DAMC, corresponding to its lower absorption capacity. These structural variations highlight the importance of pore size and surface morphology in enhancing water retention.

The SA-DAB hydrogel, which demonstrated the highest absorption capacity of 1300 g g^{-1} , also exhibited a low crystallinity index of 41% and the highest aldehyde content ($3.3 \text{ mmol} \cdot \text{g}^{-1}$ for 0.5 equivalents of periodate) among the tested DACs. These findings confirm that the cellulose source and its characteristics, such as aldehyde content and crystallinity, play a crucial role in determining the properties of hybrid SAPs, particularly their swelling capacities. By integrating chemical, structural, and morphological evidence, this study provides a comprehensive understanding of the factors influencing the performance of DAC-based hydrogels.

The absorbance capacity of 1300 g g^{-1} positions the SA-DAB hydrogel among the top-performing cellulose based hybrid SAPs. Compared to SAPs based cellulose materials reported in literature (Table 5), which typically show absorbance capacities 750 g g^{-1} (Wang et al. 2017), and synthetic options like PAA (407 g g^{-1}) (Cruz et al. 2020), the SA-DAB hydrogel demonstrates a significant enhancement. While advanced synthetic composites such as PAM/PAA/Ca (OH)₂ can achieve much higher capacities (up to 5580 g g^{-1}) (Lv et al. 2019). The SA-DAB hydrogel offers an environmentally friendly alternative with competitive performance. This combination of high absorbance and sustainable composition makes it particularly suitable for applications in agriculture, water retention, and eco-friendly slow-release fertilizers.

Conclusion

In summary, a series of eco-friendly hybrid superabsorbent hydrogels based on cellulose and PAA have been successfully developed. A comparative study of swelling performance in water was conducted to evaluate the impact of the source of cellulose and their dialdehydes cellulose (DAC) on the absorbent capacity of the hybrid SAP. The study involved the analysis of dialdehyde cellulose derived from banana, palm date, and microcrystalline cellulose. The findings revealed that increasing the NaIO_4 /

Table 5 Comparison of the capacity performance of SAPs based cellulose materials reported in the literature

SAPs based cellulose materials	Maximum swelling capacity (g g^{-1})	References
Amide-2,4 modified cellulose-g-poly acrylic acid	1253	Jamali and Rahmati (2025)
CMC- graft- sodium acrylate	306–326	Nasresfahani et al. (2025)
Cellulose-g-polyacrylic acid/coal gangue	681	Shi et al. (2024)
CMC-g-P(AA-AAm)/diatomite/DAP	105	Barala et al. (2025)
DACAc-g-poly(AA-IA)	1441	Bezbiz et al. (2024)
Poly NaCMC-g-poly(AA)	544	Klinpituksa and Kosaiyakanon (2017)
Starch (St)-graft-poly(acrylic acid (AA)-co-acrylamide(AM))/poly-vinyl alcohol (PVA) and cellulose nanocrystals (CNC)	921	Olad et al. (2020)
Ablmoschus manihot gum-g-poly (acrylic acid-co-2-acrylamide-2-methyl-1-propanesulfonic acid)/microcrystalline cellulose	1240	Li et al. (2022)
Poly (acrylic acid) grafted sodium carboxymethyl cellulose	420	Liu et al. (2022)
Poly (sodium acrylate-co-acrylamide) with 10% wt attapulgit	1257	Ren et al. (2024)
DAC-g-Poly (acrylic-co-itaconic acid)	1240	Boussif et al. (2024)
DAB-g-Poly (acrylic-co-itaconic acid)	1300	This work

glucose residue ratio resulted in a notable enhancement in the aldehyde content of all DACs. However, this increase was accompanied by decreased reaction yields due to the degradation of cellulose chains. Additionally, the X-ray analysis showed that MCC has a higher crystallinity compared to the pulps, and all DACs show a significant decrease of the crystallinity index as the NaIO_4 /glucose residue ratio. Conversely, the integration of the chosen DACs into SAPs markedly enhanced their capacity to absorb distilled water, with capacities increasing from approximately 490, 450, and 480 g g^{-1} for SA-MCC, SA-Palm, and SA-Ban, respectively, to 850, 1100, and 1300 g g^{-1} for SA-DAMC, SA-DAP, and SA-DAB, respectively. This improvement was attributed to the enhanced dispersibility in the polymer chain networks and hydrophilic properties of the dialdehyde celluloses. Moreover, the SA-DAB hydrogel demonstrated the highest absorption capacity in comparison to SA-DAP and SA-DAMC, suggesting that the swelling capacity of the SAP hydrogel is influenced by the cellulose source and properties. This study clearly shows how important the cellulose source and properties on the resulting DAC characteristic, and its significant influence on the SAP swelling capacity.

Acknowledgements The authors express their heartfelt gratitude to the Moroccan Ministry of Higher Education, Scientific Research and Innovation, CNRST, UM6P, and the OCP Foundation for their financial support through the APRD Research Program. Special appreciation is extended to the characterization test teams at CAC of Cadi Ayyad University for their invaluable contributions to our research work.

Author Contribution CY: investigation, data curation, writing original draft. AH, BAY, and BM: investigation; formal analysis, writing—reviewing and editing. LM: formal analysis, writing—reviewing and editing, supervision. BL: supervision, conceptualization, writing—reviewing and editing. ByH, MC, and CB: writing—reviewing and editing. KH: writing—reviewing and editing, supervision, methodology, funding acquisition, conceptualization.

Funding This project is fully supported by the funding of the Moroccan Ministry of Higher Education, Scientific Research and Innovation, CNRST, UM6P, and the OCP Foundation through the APRD Research Program.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interests The authors declare no competing interests.

Ethical approval

All ethical guidelines were followed and no human or animal experimentation was executed.

Consent for publication All authors have approved the manuscript and its publication.

References

- AFNOR (1979) Cellulose—determination of the dry matter content in pulps. (NF T12-011/A11979). <https://connect.snv.ch/en/nf-t12-011a1-1979>
- Aitbella H, Belachemi L, Merle N et al (2024) Schiff base functionalized cellulose: towards strong support-cobalt nanoparticles interactions for high catalytic performances. *Molecules* 29:1734. <https://doi.org/10.3390/MOLECULES29081734>
- Akatwijuka O, Gepreel MAH, Abdel-Mawgood A et al (2024) Overview of banana cellulosic fibers: agro-biomass potential, fiber extraction, properties, and sustainable applications. *Biomass Convers Biorefin* 14:7449–7465. <https://doi.org/10.1007/s13399-022-02819-0>
- Alotaibi MD, Alshammari BA, Saba N et al (2019) Characterization of natural fiber obtained from different parts of date palm tree (*Phoenix dactylifera* L.). *Int J Biol Macromol* 135:69–76. <https://doi.org/10.1016/J.IJBIOMAC.2019.05.102>
- Amutha K, Sudha A, Saravanan D (2022) Characterization of natural fibers extracted from banana inflorescence bracts. *J Nat Fibers* 19:872–881. <https://doi.org/10.1080/15440478.2020.1764437>
- Azzam F, Galliot M, Putaux JL et al (2015) Surface peeling of cellulose nanocrystals resulting from periodate oxidation and reductive amination with water-soluble polymers. *Cellulose* 22:3701–3714. <https://doi.org/10.1007/s10570-015-0785-x>
- Barala P, Lohchab M, Hooda V (2025) Carboxymethyl cellulose-based composite hydrogels for sustainable agriculture under drought stress. *Polym Adv Technol* 36:e70046. <https://doi.org/10.1002/PAT.70046>
- Behera S, Mahanwar PA (2020) Superabsorbent polymers in agriculture and other applications: a review. *Polym Technol Mater* 59:341–356. <https://doi.org/10.1080/25740881.2019.1647239>
- Bendahou A, Dufresne A, Kaddami H, Habibi Y (2007) Isolation and structural characterization of hemicelluloses from palm of *Phoenix dactylifera* L. *Carbohydr Polym* 68:601–608. <https://doi.org/10.1016/J.CARBPOL.2006.10.016>
- Bendahou A, Habibi Y, Kaddami H, Dufresne A (2009) Physico-chemical characterization of palm from *Phoenix dactylifera* L., preparation of cellulose whiskers and natural rubber-based nanocomposites. *J Biobased Mater Bioenergy* 3:81–90. <https://doi.org/10.1166/jbmb.2009.1011>

- Benhamou K, Kaddami H, Magnin A et al (2015) Bio-based polyurethane reinforced with cellulose nanofibers: a comprehensive investigation on the effect of interface. *Carbohydr Polym* 122:202–211. <https://doi.org/10.1016/J.CARPOL.2014.12.081>
- Benhamou K, Kaddami H, Dufresne A et al (2018) Impact of TEMPO-oxidation strength on the properties of cellulose nanofibril reinforced polyvinyl acetate nanocomposites. *Carbohydr Polym* 181:1061–1070. <https://doi.org/10.1016/J.CARPOL.2017.11.043>
- Beroual M, Trache D, Mehelli O et al (2021) Effect of the delignification process on the physicochemical properties and thermal stability of microcrystalline cellulose extracted from date palm fronds. *Waste Biomass Valoriz* 12:2779–2793. <https://doi.org/10.1007/S12649-020-01198-9>
- Bezbiz M, Boussif AY, Latifi M et al (2024) New polysaccharide-based bio-super-absorbents for water stress control in agriculture: preparation and properties. *EuroMediterr J Env Integr*. <https://doi.org/10.1007/s41207-024-00596-y>
- Bora A, Karak N (2022) Starch and itaconic acid-based superabsorbent hydrogels for agricultural application. *Eur Polym J* 176:111430. <https://doi.org/10.1016/J.EURPOLYMJ.2022.111430>
- Boussif AY, Bezbiz M, Belachemi L et al (2024) Preparation of superabsorbent composite(s) based on dialdehyde cellulose extracted from banana fiber waste. *Carbohydr Polym*. <https://doi.org/10.1016/j.carbpol.2024.122504>
- Călina I, Demeter M, Scărișoreanu A, Micutz M (2021) Development of novel superabsorbent hybrid hydrogels by E-beam crosslinking. *Gels* 7:189. <https://doi.org/10.3390/GELS7040189>
- Chavan VD, Pinjari DV, Waghmare NG et al (2024) A critical review on technological development of cellulosic material as a sustainable alternative for superabsorbent polymers and its recent applications. *Chem Eng J* 499:156487. <https://doi.org/10.1016/J.CEJ.2024.156487>
- Chen D, van de Ven TGM (2016) Morphological changes of sterically stabilized nanocrystalline cellulose after periodate oxidation. *Cellulose* 23:1051–1059. <https://doi.org/10.1007/S10570-016-0862-9>
- Chen J, Zhang W, Li X (2015) Preparation and characterization of a novel superabsorbent of konjac glucomannan-poly(acrylic acid) with trimethylolpropane trimethacrylate cross-linker. *RSC Adv* 5:38417–38423. <https://doi.org/10.1039/C5RA04522C>
- Cruz H, Laycock B, Strounina E et al (2020) Modified poly(acrylic acid)-based hydrogels for enhanced mainstream removal of ammonium from domestic wastewater. *Environ Sci Technol* 54:9573–9583. <https://doi.org/10.1021/ACS.EST.9B07032>
- Dang X, Liu P, Yang M et al (2019) Production and characterization of dialdehyde cellulose through green and sustainable approach. *Cellulose* 26:9503–9515. <https://doi.org/10.1007/S10570-019-02747-9>
- de Castro SC, Stanisic D, Tasic L (2024) Sequential extraction of hesperidin, pectin, lignin, and cellulose from orange peels: towards valorization of agro-waste. *Biofuel Bioprod Biorefin* 18:804–817. <https://doi.org/10.1002/BBB.2606>
- Dubinsky S, Grader GS, Shter GE, Silverstein MS (2004) Thermal degradation of poly(acrylic acid) containing copper nitrate. *Polym Degrad Stab* 86:171–178. <https://doi.org/10.1016/j.polymdegradstab.2004.04.009>
- El Idrissi N, Belachemi L, Merle N et al (2022) Comprehensive preparation and catalytic activities of Co/TEMPO-cellulose nanocomposites: a promising green catalyst. *Carbohydr Polym* 295:119765. <https://doi.org/10.1016/J.CARPOL.2022.119765>
- El-Sakhawy M, Kamel S, Salama A, Tohamy HAS (2018) Preparation and infrared study of cellulose based amphiphilic materials. *Cellul Chem Technol* 52:193–200
- Elseify LA, Midani M, Shihata LA, El-Mously H (2019) Review on cellulosic fibers extracted from date palms (*Phoenix dactylifera* L.) and their applications. *Cellulose* 26:2209–2232. <https://doi.org/10.1007/s10570-019-02259-6>
- Elseify LA, Midani M, El-Badawy AA et al (2023) Comparative study of long date palm (*Phoenix dactylifera* L.) midrib and spadix fibers with other commercial leaf fibers. *Cellulose* 30:1927–1942. <https://doi.org/10.1007/S10570-022-04972-1>
- Errokhi A, Magnin A, Putaux JL, Boufi S (2018) Morphology of the nanocellulose produced by periodate oxidation and reductive treatment of cellulose fibers. *Cellulose* 25:3899–3911. <https://doi.org/10.1007/S10570-018-1871-7>
- French AD (2014) Idealized powder diffraction patterns for cellulose polymorphs. *Cellulose* 21:885–896. <https://doi.org/10.1007/S10570-013-0030-4>
- Ghodake VB, Khare RA, Mhaske ST (2023) A simple approach towards tuning morphology of microcrystalline cellulose. *Cellul Chem Technol* 57:475–485. <https://doi.org/10.35812/CelluloseChemTechnol.2023.57.43>
- Hamdan MA, Ramli NA, Othman NA et al (2019) Characterization and property investigation of microcrystalline cellulose (MCC) and carboxymethyl cellulose (CMC) filler on the carrageenan-based biocomposite film. *Mater Today Proc* 42:56–62. <https://doi.org/10.1016/j.matpr.2020.09.304>
- Hosoya T, Bacher M, Potthast A et al (2018) Insights into degradation pathways of oxidized anhydroglucose units in cellulose by β -alkoxy-elimination: a combined theoretical and experimental approach. *Cellulose* 25:3797–3814. <https://doi.org/10.1007/S10570-018-1835>
- Huang Z, Liu C, Feng X et al (2020) Effect of regeneration solvent on the characteristics of regenerated cellulose from lithium bromide trihydrate molten salt. *Cellulose* 27:9243–9256. <https://doi.org/10.1007/S10570-020-03440-Y>
- Jamali F, Rahmati A (2025) Amide modified cellulose-g-poly acrylic acid as a superabsorbent for water retention and soil conditioner. *Int J Biol Macromol* 295:139558. <https://doi.org/10.1016/J.IJBIOMAC.2025.139558>
- Kim UJ, Kuga S, Wada M et al (2000) Periodate oxidation of crystalline cellulose. *Biomacromol* 1:488–492. <https://doi.org/10.1021/BM0000337>
- Klinpituksa P, Kosaiyakanon P (2017) Superabsorbent polymer based on sodium carboxymethyl cellulose grafted polyacrylic acid by inverse suspension polymerization. *Int J Polym Sci* 2017:3476921. <https://doi.org/10.1155/2017/3476921>

- Kumar V, Chakraborty P, Janghu P et al (2023) Potential of banana based cellulose materials for advanced applications: a review on properties and technical challenges. *Carbohydr Polym Technol Appl*. <https://doi.org/10.1016/j.carpta.2023.100366>
- Kwon YR, Kim JS (2021) Kim DH (2021) Effective enhancement of water absorbency of itaconic acid based-superabsorbent polymer via tunable surface—crosslinking. *Polym* 13:2782. <https://doi.org/10.3390/POLYM13162782>
- Lai TSM, Jayamani E, Soon KH (2023) Comparative study on thermogravimetric analysis of banana fibers treated with chemicals. *Mater Today Proc* 78:458–461. <https://doi.org/10.1016/J.MATPR.2022.10.267>
- Le Troedec M, Sedan D, Peyratout C et al (2008) Influence of various chemical treatments on the composition and structure of hemp fibres. *Compos Part A Appl Sci Manuf* 39:514–522. <https://doi.org/10.1016/J.COMPOSITESA.2007.12.001>
- Li X, Wang X, Sang W et al (2022) Preparation and anti-leakage performances of superabsorbent composite based on abmoschus manihot gum and microcrystalline cellulose. *J Environ Chem Eng* 10:107644. <https://doi.org/10.1016/J.JECE.2022.107644>
- Liu Y, Zhu Y, Mu B et al (2022) Synthesis, characterization, and swelling behaviors of sodium carboxymethyl cellulose-g-poly(acrylic acid)/semi-coke superabsorbent. *Polym Bull* 79:935–953. <https://doi.org/10.1007/S00289-021-03545-9>
- Lu J, Li Y, Hu D et al (2015) Synthesis and properties of pH-, thermo-, and salt-sensitive modified poly(aspartic acid)/poly(vinyl alcohol) IPN hydrogel and its drug controlled release. *BioMed Res Int*. <https://doi.org/10.1155/2015/236745>
- Lupidi G, Pastore G, Marcantoni E, Gabrielli S (2023) Recent developments in chemical derivatization of microcrystalline cellulose (MCC): pre-treatments, functionalization, and applications. *Molecules*. <https://doi.org/10.3390/molecules28052009>
- Lv Q, Wu M, Shen Y (2019) Enhanced swelling ratio and water retention capacity for novel super-absorbent hydrogel. *Colloids Surf A Physicochem Eng Asp* 583:123972. <https://doi.org/10.1016/j.colsurfa.2019.123972>
- Lv J, Zou C, Dai Q et al (2024) Pickering emulsions stabilized by aldehyde-modified cellulose nanofibrils: stabilization and asphalt recovery application. *Ind Crops Prod* 218:118967. <https://doi.org/10.1016/J.INDCROP.2024.118967>
- Nasresfahani M, Babaeipour V, Imani M (2025) Improving the water absorption properties of bacterial cellulose by in-situ and ex-situ modifications for use in CMC-graft-sodium acrylate superabsorbent. *Colloid Polym Sci* 303:759–777
- Nypelö T, Berke B, Spirk S, Sirviö JA (2021) Periodate oxidation of wood polysaccharides—modulation of hierarchies. *Carbohydr Polym*. <https://doi.org/10.1016/j.carbpol.2020.117105>
- Olad A, Doustdar F, Gharekhani H (2020) Fabrication and characterization of a starch-based superabsorbent hydrogel composite reinforced with cellulose nanocrystals from potato peel waste. *Colloids Surf A Physicochem Eng Asp* 601:124962. <https://doi.org/10.1016/J.COLSURFA.2020.124962>
- Oladosu Y, Rafii MY, Arolo F et al (2022) Superabsorbent polymer hydrogels for sustainable agriculture: a review. *Hortic* 8:605. <https://doi.org/10.3390/HORTICULTURAE8070605>
- Oyarce E, Pizarro GDC, Oyarzún DP et al (2020) Adsorption of methylene blue in aqueous solution using hydrogels based on 2-hydroxyethyl methacrylate copolymerized with itaconic acid or acrylic acid. *Mater Today Commun* 25:101324. <https://doi.org/10.1016/J.MTCOMM.2020.101324>
- Pérez-Álvarez L, Ruiz-Rubio L, Lizundia E, Vilas-Vilela JL (2018) Polysaccharide-based superabsorbents: synthesis, properties, and applications. In: Mondal M (ed) *Polymers and polymeric composites: a reference series (POPOC)*. Springer, Cham, pp 1–39
- Plappert SF, Quraishi S, Pircher N et al (2018) Transparent, flexible, and strong 2,3-dialdehyde cellulose films with high oxygen barrier properties. *Biomacromol* 19:2969–2978. <https://doi.org/10.1021/ACS.BIOMAC.8B00536>
- Putra NR, Aziz AHA, Faizal ANM, Che Yunus MA (2022) Methods and potential in valorization of banana peels waste by various extraction processes. In Review. *Sustain* 14:10571. <https://doi.org/10.3390/SU141710571>
- Raza M, Abu-Jdayil B, Banat F, Al-Marzouqi AH (2022) Isolation and characterization of cellulose nanocrystals from date palm waste. *ACS Omega* 7:25366–25379. <https://doi.org/10.1021/acsomega.2c02333>
- Ren H, Ren J, Tao L, Ren X (2024) Swelling behaviors of superabsorbent composites based on acrylic acid/acrylamide copolymer and acid-modified attapulgite. *RSC Adv* 14:14758–14766. <https://doi.org/10.1039/D4RA01021C>
- Ruan C, Strømme M, Lindh J (2016) A green and simple method for preparation of an efficient palladium adsorbent based on cysteine functionalized 2,3-dialdehyde cellulose. *Cellulose* 23:2627–2638. <https://doi.org/10.1007/s10570-016-0976-0>
- Ruan CQ, Strømme M, Lindh J (2018) Preparation of porous 2,3-dialdehyde cellulose beads crosslinked with chitosan and their application in adsorption of Congo red dye. *Carbohydr Polym* 181:200–207. <https://doi.org/10.1016/J.CARBPOL.2017.10.072>
- Saraswathy M, Komath M, Ragini DD et al (2023) Bactericidal activity of superabsorbent polymer granules for their applications in respiratory fluid solidification systems. *ACS Omega* 8:25114–25121. <https://doi.org/10.1021/ACSOMEGA.3C01994>
- Saud A, Saleem H, Khan AW et al (2023) Date palm tree leaf-derived cellulose nanocrystal incorporated thin-film composite forward osmosis membranes for produced water treatment. *Membranes (Basel)*. <https://doi.org/10.3390/membranes13050513>
- Sayyed AJ, Mohite LV, Deshmukh NA, Pinjari DV (2018) Effect of ultrasound treatment on swelling behavior of cellulose in aqueous N-methyl-morpholine-N-oxide solution. *Ultrason Sonochem* 49:161–168. <https://doi.org/10.1016/J.ULTSONCH.2018.07.042>
- Sayyed AJ, Pinjari DV, Sonawane SH et al (2021) Cellulose-based nanomaterials for water and wastewater treatments:

- a review. *J Environ Chem Eng* 9:106626. <https://doi.org/10.1016/J.JECE.2021.106626>
- Sbiai A, Kaddami H, Sautereau H et al (2011) TEMPO-mediated oxidation of lignocellulosic fibers from date palm leaves. *Carbohydr Polym* 86:1445–1450. <https://doi.org/10.1016/J.CARPOL.2011.06.005>
- Segal L, Creely JJ, Martin AE, Conrad CM (1959) An empirical method for estimating the degree of crystallinity of native cellulose using the X-ray diffractometer. *Text Res J* 29:786–794. <https://doi.org/10.1177/004051755902901003>
- Sharma M, Bains A, Goksen G et al (2024) A review of valorization of agricultural waste for the synthesis of cellulose membranes: separation of organic, inorganic, and microbial pollutants. *Int J Biol Macromol* 277:134170. <https://doi.org/10.1016/J.IJBIOMAC.2024.134170>
- Shi S, Yan, Y, Wang, X-X, Zhang X, Ma Y, Gao P, Zhang Y (2024) Fabrication and application of eco-friendly superabsorbent composites based on cellulose and coal gangue. <https://doi.org/10.2139/SSRN.5057133>
- Siller M, Amer H, Bacher M et al (2015) Effects of periodate oxidation on cellulose polymorphs. *Cellulose* 22:2245–2261. <https://doi.org/10.1007/S10570-015-0648-5>
- Simon J, Tsetsge O, Iqbal NA et al (2022) A fast method to measure the degree of oxidation of dialdehyde celluloses using multivariate calibration and infrared spectroscopy. *Carbohydr Polym*. <https://doi.org/10.1016/j.carbpol.2021.118887>
- Song J, Zhao H, Zhao G et al (2019) Novel semi-IPN nanocomposites with functions of both nutrient slow-release and water retention. 1. Microscopic structure, water absorbency, and degradation performance. *J Agric Food Chem* 67:7587–7597. <https://doi.org/10.1021/ACS.JAFC.9B00888>
- Sun X, Jiang F (2024) Periodate oxidation-mediated nanocelluloses: preparation, functionalization, structural design, and applications. *Carbohydr Polym* 341:122305. <https://doi.org/10.1016/J.CARPOL.2024.122305>
- Sun B, Zhang M, Hou Q et al (2016) Further characterization of cellulose nanocrystal (CNC) preparation from sulfuric acid hydrolysis of cotton fibers. *Cellulose* 23:439–450. <https://doi.org/10.1007/S10570-015-0803-Z>
- Takahashi M, Kosaka I, Ohta S (2023) Water retention characteristics of superabsorbent polymers (SAPs) used as soil amendments. *Soil Syst*. <https://doi.org/10.3390/soilsystem57020058>
- Teleky BE, Vodnar DC (2021) Recent advances in biotechnological itaconic acid production, and application for a sustainable approach. *Polym* 13:3574. <https://doi.org/10.3390/POLYM13203574>
- Ventura-Cruz S, Flores-Alamo N, Tecante A (2020) Preparation of microcrystalline cellulose from residual Rose stems (*Rosa* spp.) by successive delignification with alkaline hydrogen peroxide. *Int J Biol Macromol* 155:324–329. <https://doi.org/10.1016/j.ijbiomac.2020.03.222>
- Wang Z, Ning A, Xie P et al (2017) Synthesis and swelling behaviors of carboxymethyl cellulose-based superabsorbent resin hybridized with graphene oxide. *Carbohydr Polym* 157:48–56. <https://doi.org/10.1016/J.CARPOL.2016.09.070>
- Wegrzynowska-Drzymalska K, Mlynarczyk DT, Chelminiak-Dudkiewicz D et al (2022) Chitosan-gelatin films cross-linked with dialdehyde cellulose nanocrystals as potential materials for wound dressings. *Int J Mol Sci* 23:9700. <https://doi.org/10.3390/IJMS23179700>
- Yang Y, Liang Z, Zhang R et al (2024) Research advances in superabsorbent polymers. *Polymers* 16:10.3390/polym16040501
- Yu Y, Guo W, Qu J et al (2023) Preparation and characterization of dialdehyde cellulose nanocrystals from the waste nutshell. *Environ Dev Sustain*. <https://doi.org/10.1007/S10668-023-04332-4>
- Zhang Y, Wei J, Zhang X, Gao P (2023) Fabrication and swelling properties of a novel superabsorbent composite derived from waste coal gangue. *Polym Polym Compos*. <https://doi.org/10.1177/09673911231182902>
- Zou Y, Zhang A, Lin L et al (2023) Schiff base cross-linked dialdehyde cellulose/gelatin composite aerogels as porous structure templates for oleogels preparation. *Int J Biol Macromol* 224:667–675. <https://doi.org/10.1016/J.IJBIOMAC.2022.10.155>
- Zuluaga R, Putaux JL, Cruz J et al (2009) Cellulose microfibrils from banana rachis: effect of alkaline treatments on structural and morphological features. *Carbohydr Polym* 76:51–59. <https://doi.org/10.1016/J.CARPOL.2008.09.024>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.